

Reference

```
def fr(xr_k, thr_k):  
    ...  
    return xr_next  
  
def hr(xr_k, thr_k):  
    ...  
    return s_k  
  
ref_signal = DynaSys(f=fr,h=hr)
```

Controller

```
def fc(xc_k, s_k, xhatp_k, thc_k):  
    ...  
    return xc_next  
  
def hc(xc_k, s_k, xhatp_k, thc_k):  
    ...  
    return u_k  
  
controller = DynaSys(f=fc,h=hc)
```

Plant

```
def fp(xp_k, u_k, thp_k):  
    ...  
    return xp_next  
  
def hp(xp_k, u_k, thp_k):  
    ...  
    return y_k  
  
plant = DynaSys(f=fp,h=hp)
```

Observer

```
def fo(xo_k, u_k, y_k, tho_k):  
    ...  
    return xo_next  
  
def ho(xo_k, u_k, y_k, tho_k):  
    ...  
    return xhatp_k  
  
observer = DynaSys(f=fo,h=ho)
```